

2.15 ▲ (Optional) — Is $\lim_{x \rightarrow c} f'(x)$ Equal to $f'(c)$?

Consider the function

$$f(x) = \begin{cases} \frac{\sin x^2}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

For any $x \neq 0$ we can easily use our differentiation rules to find

$$f'(x) = \frac{2x^2 \cos x^2 - \sin x^2}{x^2}$$

But for $x = 0$ none of our usual differentiation rules apply. So how do we find $f'(0)$? One obviously legitimate strategy is to directly apply the Definition 2.2.1 of the derivative. As an alternative, we will now consider the question “Can one find $f'(0)$ by taking the limit of $f'(x)$ as x tends to zero?”. There is bad news and there is good news.

- The bad news is that, even for functions $f(x)$ that are differentiable for all x , $f'(x)$ need not be continuous. That is, it is *not* always true that $\lim_{x \rightarrow 0} f'(x) = f'(0)$. We will see a function for which $\lim_{x \rightarrow 0} f'(x) \neq f'(0)$ in Example 2.15.1 below.
- The good news is that Theorem 2.15.2 below provides conditions which are sufficient to guarantee that $f(x)$ is differentiable at $x = 0$ and that $\lim_{x \rightarrow 0} f'(x) = f'(0)$.

Example 2.15.1

Consider the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

For $x \neq 0$ we have, by the product and chain rules,

$$\begin{aligned} f'(x) &= 2x \sin \frac{1}{x} + x^2 \left(\cos \frac{1}{x} \right) \left(-\frac{1}{x^2} \right) \\ &= 2x \sin \frac{1}{x} - \cos \frac{1}{x} \end{aligned}$$

As $\left| \sin \frac{1}{x} \right| \leq 1$, we have

$$\lim_{x \rightarrow 0} 2x \sin \frac{1}{x} = 0$$

On the other hand, as x tends to zero, $\frac{1}{x}$ goes to $\pm\infty$. So

$$\lim_{x \rightarrow 0} \cos \frac{1}{x} = DNE \implies \lim_{x \rightarrow 0} f'(x) = DNE$$

We will now see that, despite this, $f'(0)$ is perfectly well defined. By definition

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h} \\ &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} \\ &= 0 \quad \text{since } \left| \sin \frac{1}{h} \right| \leq 1 \end{aligned}$$

So $f'(0)$ exists, but is not equal to $\lim_{x \rightarrow 0} f'(x)$, which does not exist.

Example 2.15.1

Now for the good news.

Theorem 2.15.2.

Let $a < c < b$. Assume that

- the function $f(x)$ is continuous on the interval $a < x < b$ and
- is differentiable at every x in the intervals $a < x < c$ and $c < x < b$ and
- the limit $\lim_{x \rightarrow c} f'(x)$ exists.

Then f is differentiable at $x = c$ and

$$f'(c) = \lim_{x \rightarrow c} f'(x)$$

Proof. By hypothesis, there is a number L such that

$$\lim_{x \rightarrow c} f'(x) = L$$

By definition

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

By the Mean Value Theorem (Theorem 2.13.4) there is, for each h , an (unknown) number x_h between c and $c+h$ such that $f'(x_h) = \frac{f(c+h) - f(c)}{h}$. So

$$f'(c) = \lim_{h \rightarrow 0} f'(x_h)$$

As h tends to zero, $c+h$ tends to c , and so x_h is forced to tend to c , and $f'(x_h)$ is forced to tend to L so that

$$f'(c) = \lim_{h \rightarrow 0} f'(x_h) = L$$

□

In the next example we evaluate $f'(0)$ by applying Theorem 2.15.2

Example 2.15.3

Let

$$f(x) = \begin{cases} \frac{\sin x^2}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

We have already observed above that, for $x \neq 0$,

$$f'(x) = \frac{2x^2 \cos x^2 - \sin x^2}{x^2} = 2 \cos x^2 - \frac{\sin x^2}{x^2}$$

We use Theorem 2.15.2 with $c = 0$ to show that $f(x)$ is differentiable at $x = 0$ and to evaluate $f'(0)$. That theorem has two hypotheses that we have not yet verified, namely the continuity of $f(x)$ at $x = 0$, and the existence of the limit $\lim_{x \rightarrow 0} f'(x)$. We verify them now.

- We already know, by Lemma 2.8.1 that $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$. So

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} &= \lim_{h \rightarrow 0^+} \frac{\sin h}{h} \quad \text{with } h = x^2 \\ &= 1 \end{aligned}$$

and

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x^2}{x} = \lim_{x \rightarrow 0} x \frac{\sin x^2}{x^2} = \lim_{x \rightarrow 0} x \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = 0 \times 1 = 0$$

and $f(x)$ is continuous at $x = 0$.

- The limit of the derivative is

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \left[2 \cos x^2 - \frac{\sin x^2}{x^2} \right] = 2 \times 1 - 1 = 1$$

So, by Theorem 2.15.2, $f(x)$ is differentiable at $x = 0$ and $f'(0) = 1$.

Example 2.15.3